CS135 HW3

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Problem 1. Model creation and training

1A: Summary Statistics

- 8.7
- the accuracy of the baseline that always predicts negative will be 13110 / (13110 + 1140) = 0.92 or 92
- this model is not good enough for our readmission risk task because the percentage of true positive rate is 0 which will not be useful for us since it fails to predict patients who actually are in the risk of being readmitted to the hospital in 30 days.

1B: Logistic Regression

- best index: 12
- best c: 0.3593813663804626
- best k: 1
- best score: 0.9661368334022323

1C: K-nearest Neighbors

- - For KNN:
  - Best classifier n_neighbors: 23
  - Best imputation n_neighbors: 1
  - Validation AUCROC: 0.8193840429929723

Problem 2. Choosing a decision threshold from binary classification metrics

2A: ROC Curves

The red curve, which is the logistic regression, performs better than KNN model because it is much more close to the ideal ROC curve. Logistic regression generally performs better except at the start and finish where they align closely.

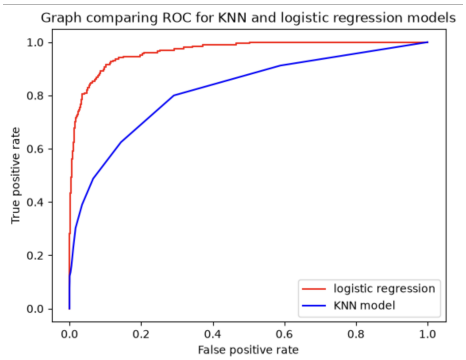


Figure 1: Graph comparing the ROC score for KNN and logistic regression models

2B: Selecting a decision threshold

- FNR: 0.15121951219512197
- TPR: 0.848780487804878
- FPR: 0.05847457627118644
- Threshold: 0.13618693960438685

- FNR: 0.19999999999999996
- TPR: 0.8
- FPR: 0.29152542372881357
- Threshold: 0.08695652173913043

2C: Test-Set Performance

- - True negatives (TN): 1140
- False positives (FP): 171
- False negatives (FN): 5
- True positives (TP): 109

Problem 3. Choosing a decision threshold based on cost

3A: Baseline cost

- excessive costs for only positive: 399999999.9999999
- excessive costs for only negatives: 14576271.186440678

3B: Calculating Total Cost

The logistic regression model is better because it has a lower extrapolated cost of 1.6 million compared to 4.2 million for the KNN model. Comparing to the 39 million for predicting all positives, this model is significantly better at lowering the extrapolated cost. However, comparing to the baseline for which it only predicts negatives which has a cost of just 1.46 million, this is more expensive. Although the baseline negatives seem like it is lowering the cost, it is potentially worse because it is sending people that needed to be readmitted back. I think the threshold of 6.7

3C: Changing Base Rates

The optimal thresholds increased from previous. Higher thresholds predict fewer patients as positive, reducing both TPR and FPR. The reason is that readmissions are now less common: there are fewer positive patients to miss and more negative patients who could receive false alarms. Consequently, reducing false positives becomes more valuable relative to catching positives, even though each false negative still costs much more. At any fixed threshold, the validation TPR and FPR stay the same in this simulation. Changing the base rate changes how those rates contribute to cost, which can change the selected threshold.